بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

In the name of Allah, Most Gracious, Most Merciful

CSE 4107
Structured Programming I

Topic: Program Organization

As Will Rgers would have said, "There is no such thing as free variable."

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Local Variables

A variable declared in the body of a function is said to be local in that function

Follow the white board

```
int sum_digits ( int n){
     int sum = 0              //Local variables
     while (n > 0 ){
          sum += n % 10
          n /= 10;
     }
     return sum;
}
```

- Automatic storage duration: storage duration/ extent of a variable is the portion of program execution during which storage for the variable exist
- Block scope: Portion of the program text in which the variable can be referenced. A local variable has block scope.

Source: C Programming Modern Approach Book

Static Local Variables

static word before the variable declaration caues it to be static storage.

Follow the white board

```
void f(void){
     static int i;      //static local variable.
}
```

- Has parmanent storage through out the execution of the program.
- Has same memory location through out the execution of the program.
- Local static still has its block scope.
- Parameters have the same properties: automatic storage and block scope

Source: C Programming Modern Approach Book

External Variables

Variables that are declared outside the body of any function. Different from local varibales.
AKA Global Variable

Follow the white board

- Static storage duration: variable life time is the life time of the program execution.
- File scope: Has its visibility from the point of declaration to the end of the enclosing file.
- Example: Implementation of stack data structure:

Pros of External Variables: Convenient when many function must share a variable or large number of variables.

Cons:

- Chaning a external variable needs recheck of every place where it was used
- If incorrect value is assigned, its difficult to find the culprit function.
- Hinders the reusability property of function.

Source: C Programming Modern Approach Book

Example Program

Using external variable in Program

Follow the white board

Source: C Programming Modern Approach Book

Blocks and Scope

The compound statement that we have seen before.

```
Follow the white board
```

Scope rule: When a declaration inside a block names as identifier that's already visible, the new declaration temporarily hides the old one. After the end of block scope, the variable regains its old value.

Source: C Programming Modern Approach Book

Organizing a C program

Strategy for arranging major element of a C program.

Follow the white board

A C program may contains the followings:

- Preprossors
- Type definitions
- Declaration of external variables
- Function prototypes
- Function definitions

Few rules:

- preprocessor doesn't take effect until the line where it appears,
- Type name can't be used untill declaration.
- Variables can't be used untill it's declaration.
- Function should be declared or defined before its' first call

Source: C Programming Modern Approach Book


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THANK
YOU